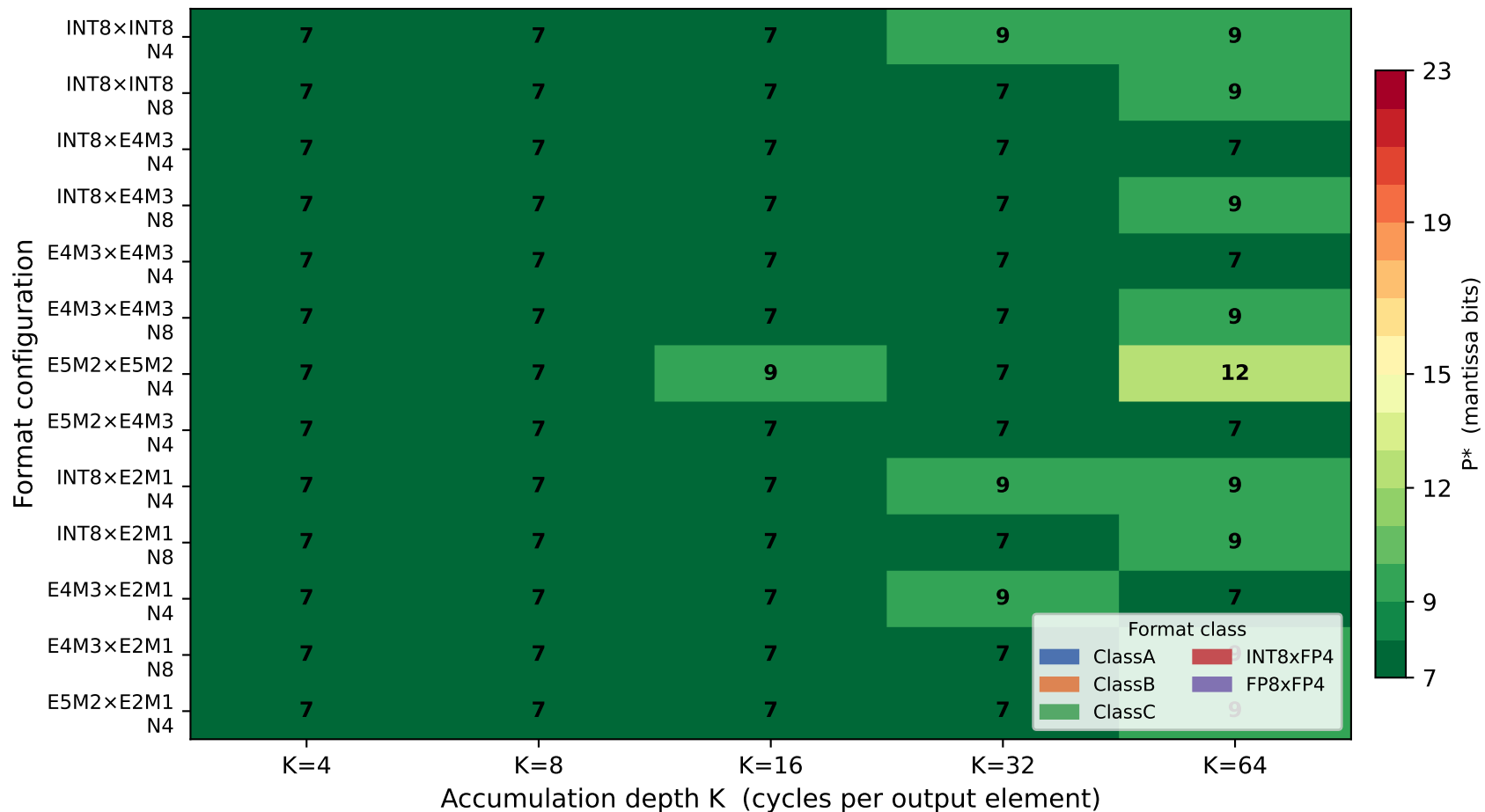


Fig 1 — Minimum sufficient accumulator mantissa bits P^*
 (normErr criterion: full-pipeline error $\leq 110\%$ of FP32 baseline)



Note: $P^*=7 \rightarrow$ BF16-equivalent mantissa. $P^*=23 \rightarrow$ full FP32. FP4 configs (E2M1) show no additional precision penalty vs FP8 counterparts — their wider requant noise floor absorbs accumulation error.